มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 4 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

Unless allowed by Mahidol University, the User shall not copy, modify, or exploit in part or in whole of the copyrighted materials on the Platform, other than for your own individual study. The copy, modify or communication to public of the copyrighted materials shall be inflicted with a fine from 20,000 Baht up to 200,000 Baht. If the offence is committed with the commercial purpose, the offender shall be inflicted with imprisonment for a term from 6 months up to 4 years or a fine from 100,000 Baht up to 800,000 Baht or both imprisonment and fine.

Mahidol University
Faculty of Information
and Communication Technology



ITDS261 Introduction to Software Engineering

Lecture 3 – Requirement Engineering

Topics

- Overview of software requirements
- Requirements engineering processes
  - Requirements elicitation
  - Requirements specification
  - Requirements validation
  - Requirements change

Overview of software requirements

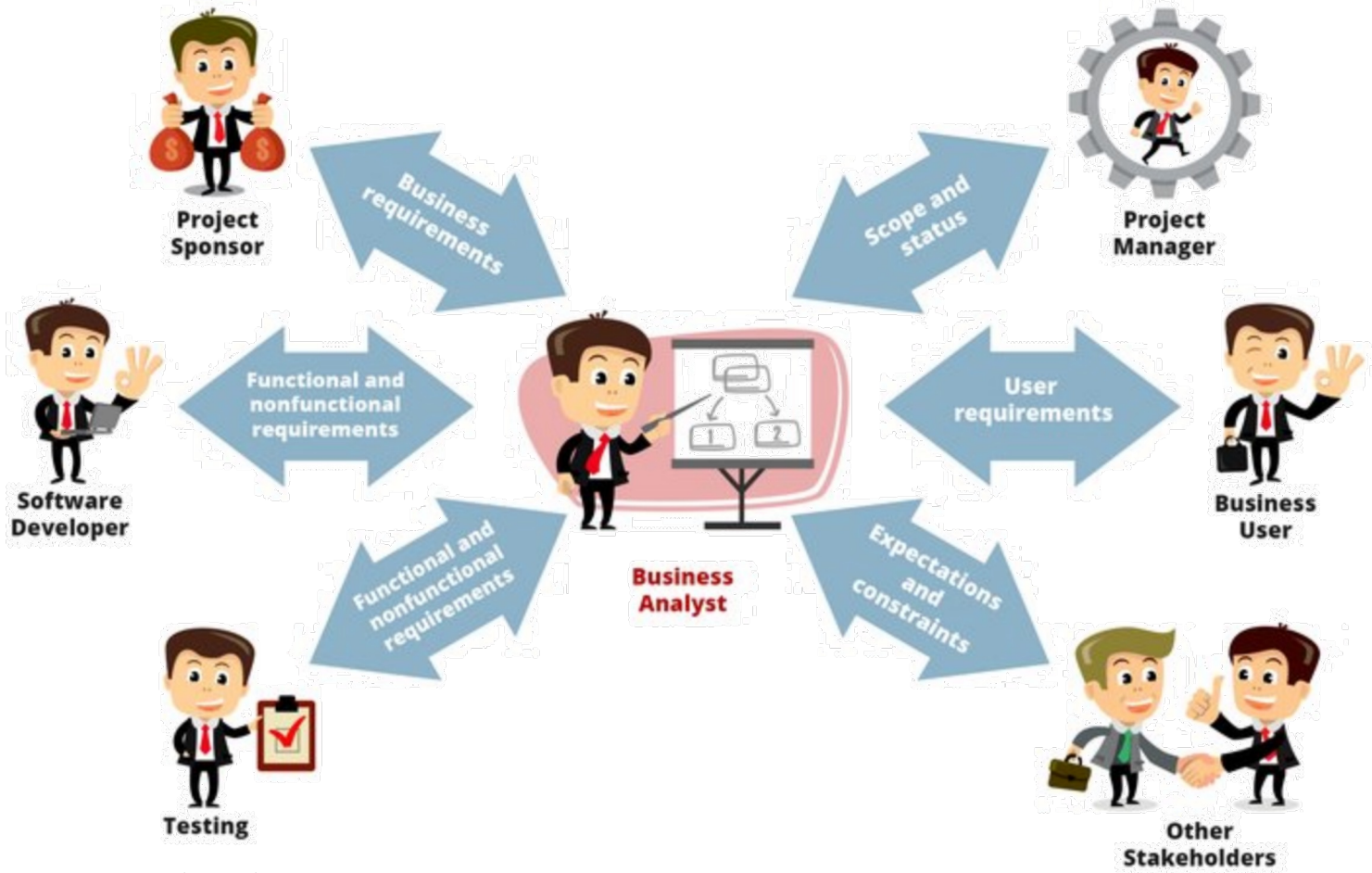
What are software requirements?

- Software requirements are the **descriptions of the services** that a system should provide and the **constraints** on its operations.
- Requirements reflect the needs of customers for a system that serves a certain purpose.
- They usually contain detailed, formal definitions of system functions.



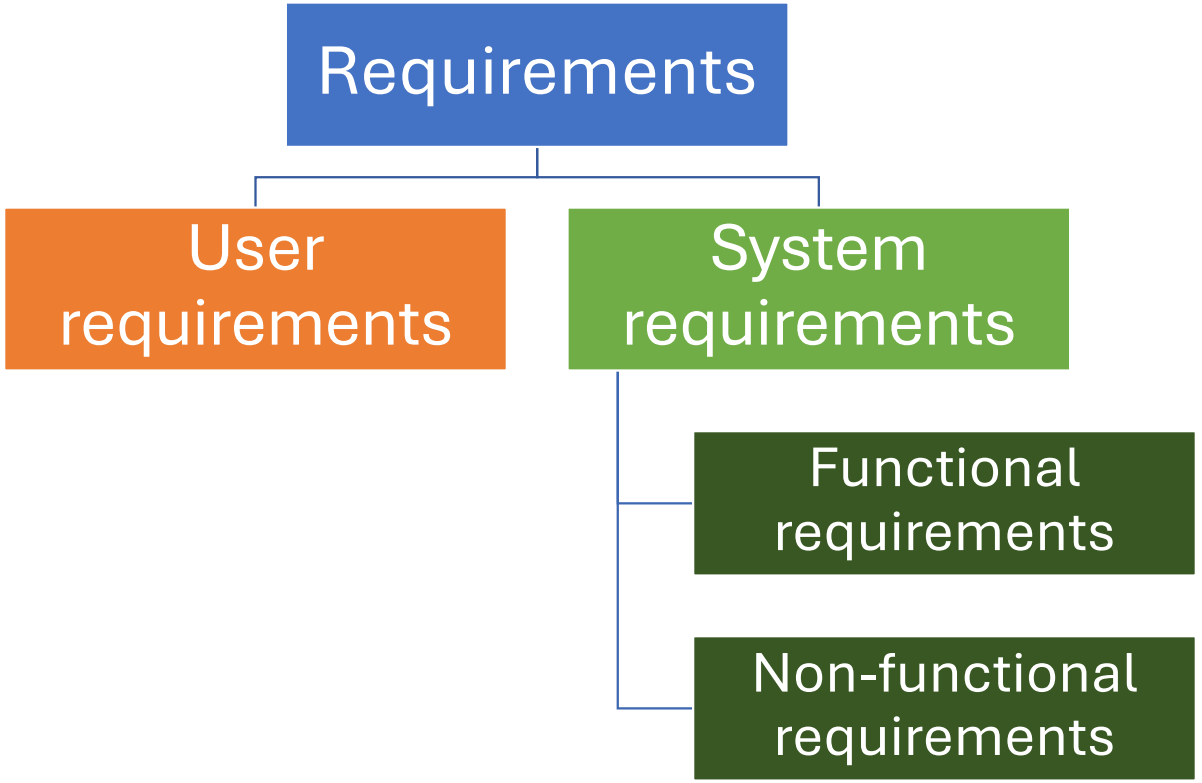
This Photo by Unknown Author is licensed under CC BY-SA

Sources of software requirements



https://www.scnsoft.com/blog/meeting-business-analyst

Types of requirements



Types of requirements

User requirements

- State what services are expected to provide to system users and the constraints under which it must operate
- May vary from broad statements of the system features required to detailed precise descriptions of the system functionality

System requirements

- State more detailed descriptions of the software system's functions, services and operational constraints
- Define exactly what is to be implemented

Example of user and system requirements

Mental health care patient information system (Mentcare)

User requirements definition

1. The Mentcare system shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

System requirements specification

1.1 On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.
1.2 The system shall generate the report for printing after 17.30 on the last working day of the month.
1.3 A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.
1.4 If drugs are available in different dose units (e.g. 10mg, 20mg, etc.) separate reports shall be created for each dose unit.
1.5 Access to drug cost reports shall be restricted to authorized users as listed on a management access control list.

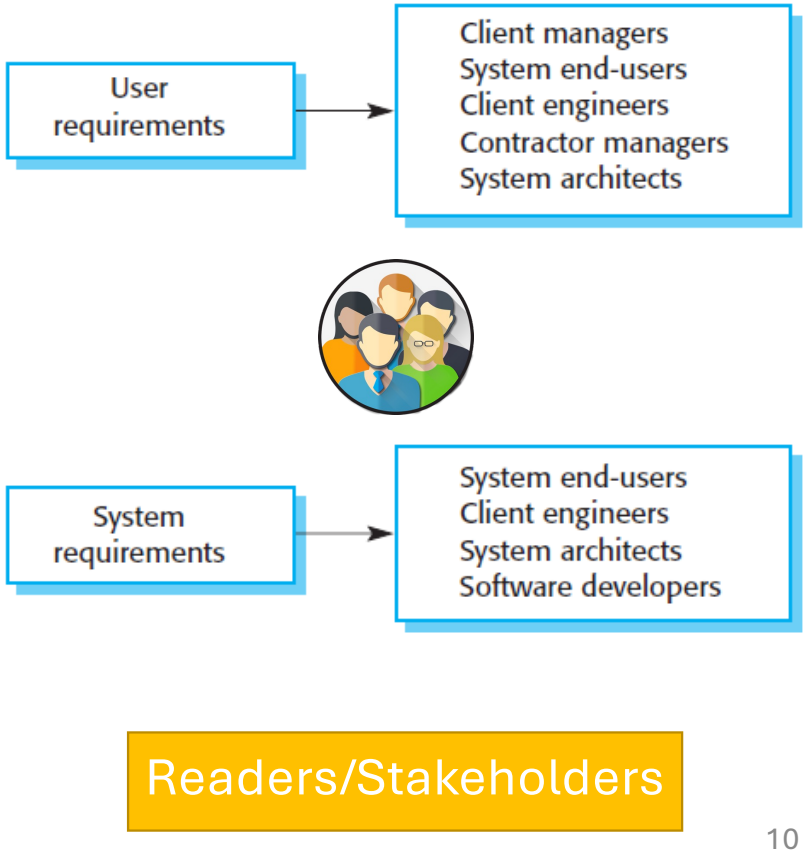
Who are expected to read those requirements?

User requirements definition

1. The Mentcare system shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

System requirements specification

1.1 On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.
1.2 The system shall generate the report for printing after 17.30 on the last working day of the month.
1.3 A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.
1.4 If drugs are available in different dose units (e.g. 10mg, 20mg, etc.) separate reports shall be created for each dose unit.
1.5 Access to drug cost reports shall be restricted to authorized users as listed on a management access control list.



Readers/Stakeholders

Example of system stakeholders in Mentcare




Patients



Doctors



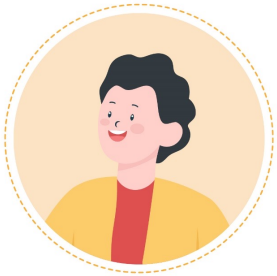
Nurses



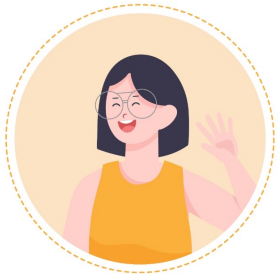

IT staff



Medical receptionists



Health care managers



Medical records staff



Medical ethics manager

Functional vs Non-functional requirements

Functional requirements

- State the services that the system should provide
- Describe that the system should do
- State how the system should react to particular inputs
- State how the system should behave in particular situations

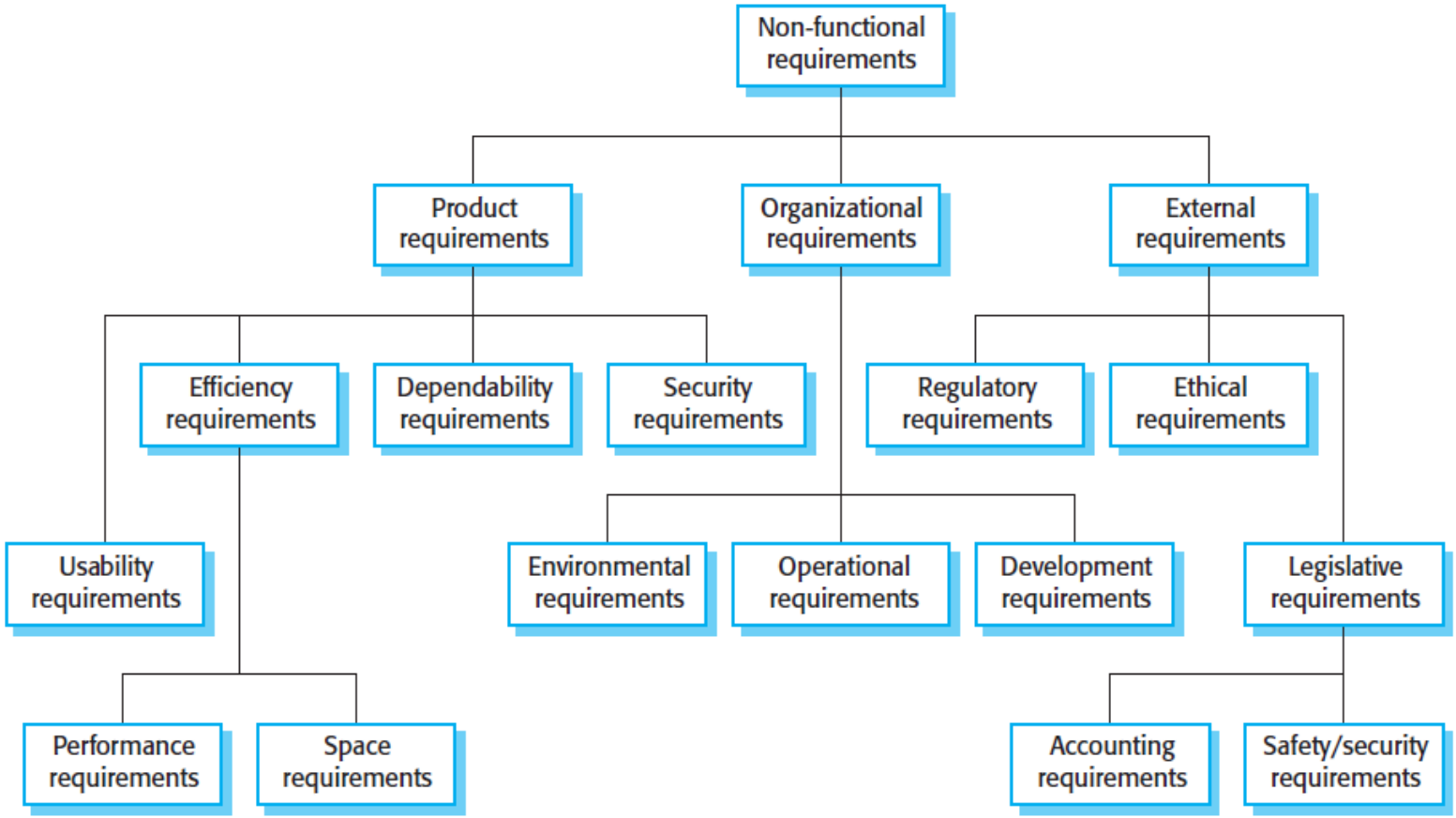
Non-functional requirements

- State the constraints on the services or functions offered by the system (e.g., response time, compatibility or memory use)
- Often apply to the system as a whole rather than individual system features or services

Example of functional requirements

- A user shall be able to search the appointments lists for all clinics.
- The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
- Each staff member using the system shall be uniquely identified by his or her eight-digit employee number.

Types of non-functional requirements



Example of non-functional requirements

Property	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Megabytes/Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

Must be measurable and testable!!!

Functional or non-functional?

The application should be easy-to-use.

The system shall allow users to create accounts.

The system shall allow users to add products to a shopping cart.

The system should be able to handle up to 10,000 concurrent users without significant performance degradation.

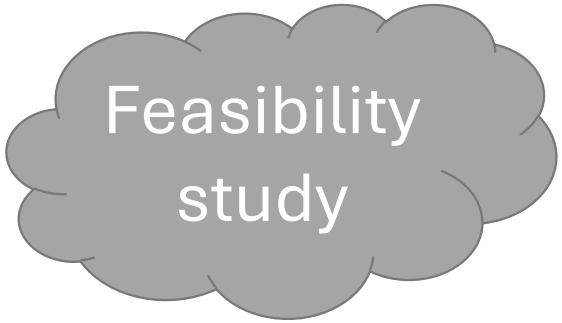
Users shall be able to search for products by keywords.

The website should load within 2 seconds for 95% of user requests.

Requirements engineering processes

The process of finding out, analysing, documenting and checking these software requirements is called ...





Requirements Engineering (RE)

Feasibility study

- A feasibility study is a short, focused study that should take place early in the RE process.
- It aims to answer these questions:
  1) Does the system contribute to the overall objectives of the organization?
  2) Can the system be implemented within schedule and budget using current technology? and
  3) Can the system be integrated with other systems that are used?

If the answer to any of these questions is no, you should probably not go ahead with the project.



Requirements engineering processes

- RE is an iterative process
- Output: System requirements document
- **4 processes:**
  - Requirements elicitation
  - Requirements specification
  - Requirements validation
  - Requirements change

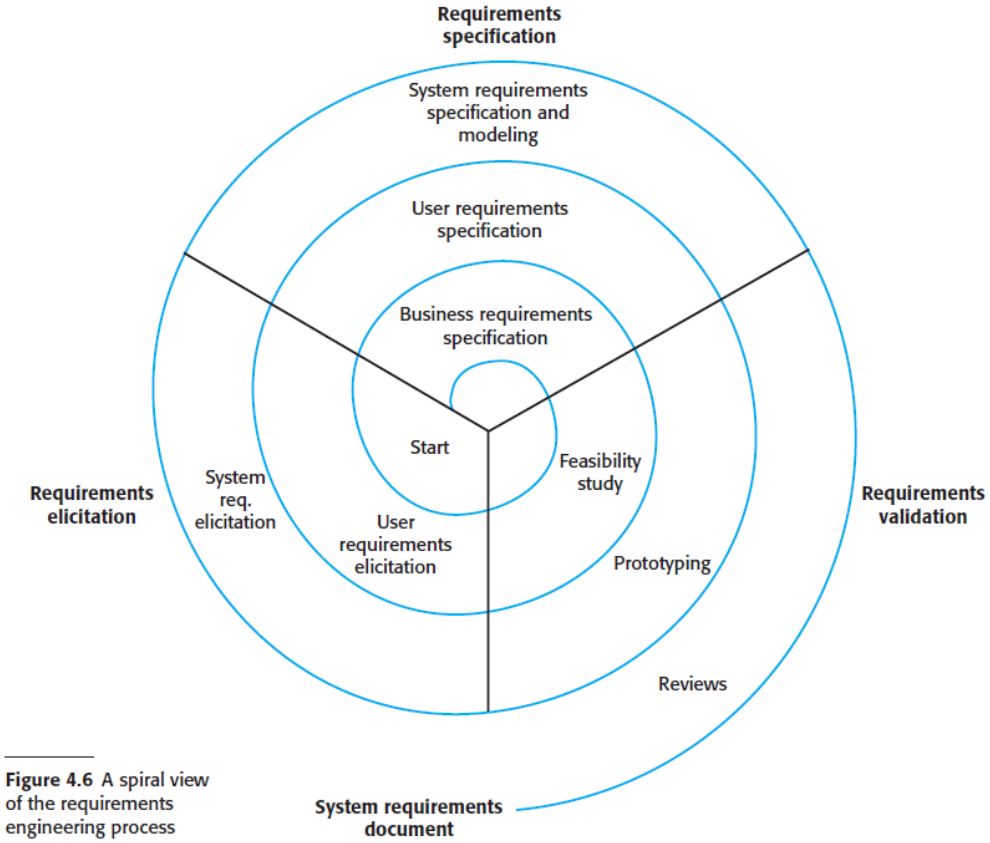


Figure 4.6 A spiral view of the requirements engineering process

Requirements elicitation

Requirements elicitation

- **Aim:** to understand the work that stakeholders do and how they might use a new system to help support that work.
- **What needs to be done?:** software engineers work with stakeholders to find out about the application domain, work activities, the services and system features that stakeholders want, the required performance of the system, hardware constraints, and so on.

Requirements elicitation challenges

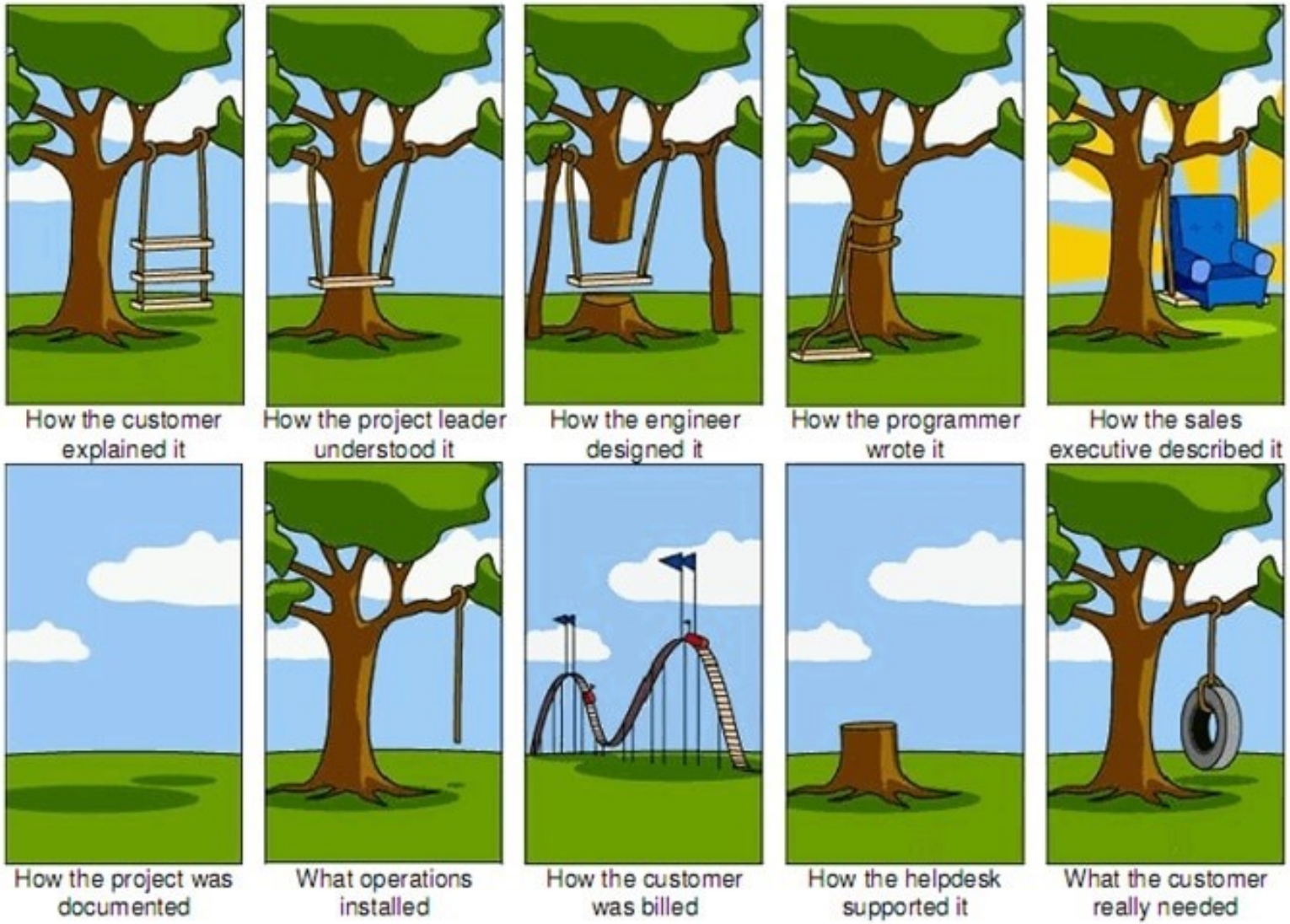
- Eliciting and understanding requirements from system stakeholders is a difficult process for several reasons:

Stakeholders often **do not know** what they want

Stakeholders may **find it difficult to articulate** what they want the system to do

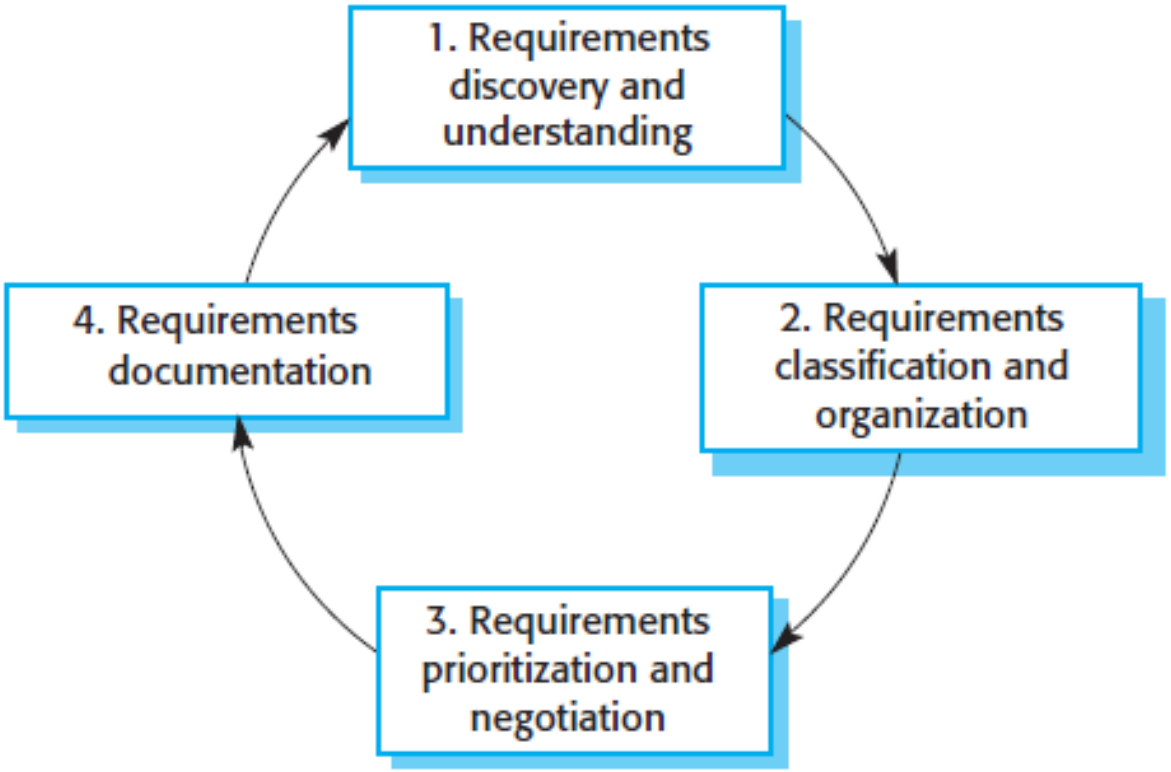
Stakeholders may **make unrealistic demands** because they do not know what is and isn't feasible

Different stakeholders, with diverse requirements, may **express their requirements in different ways**



Who should be the key man to facilitate the requirements elicitation?

Requirements elicitation and analysis process



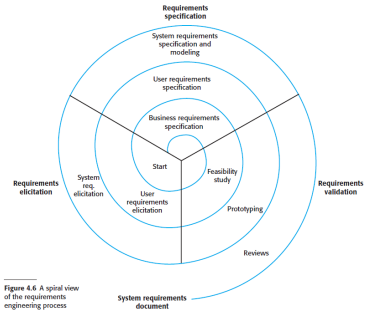
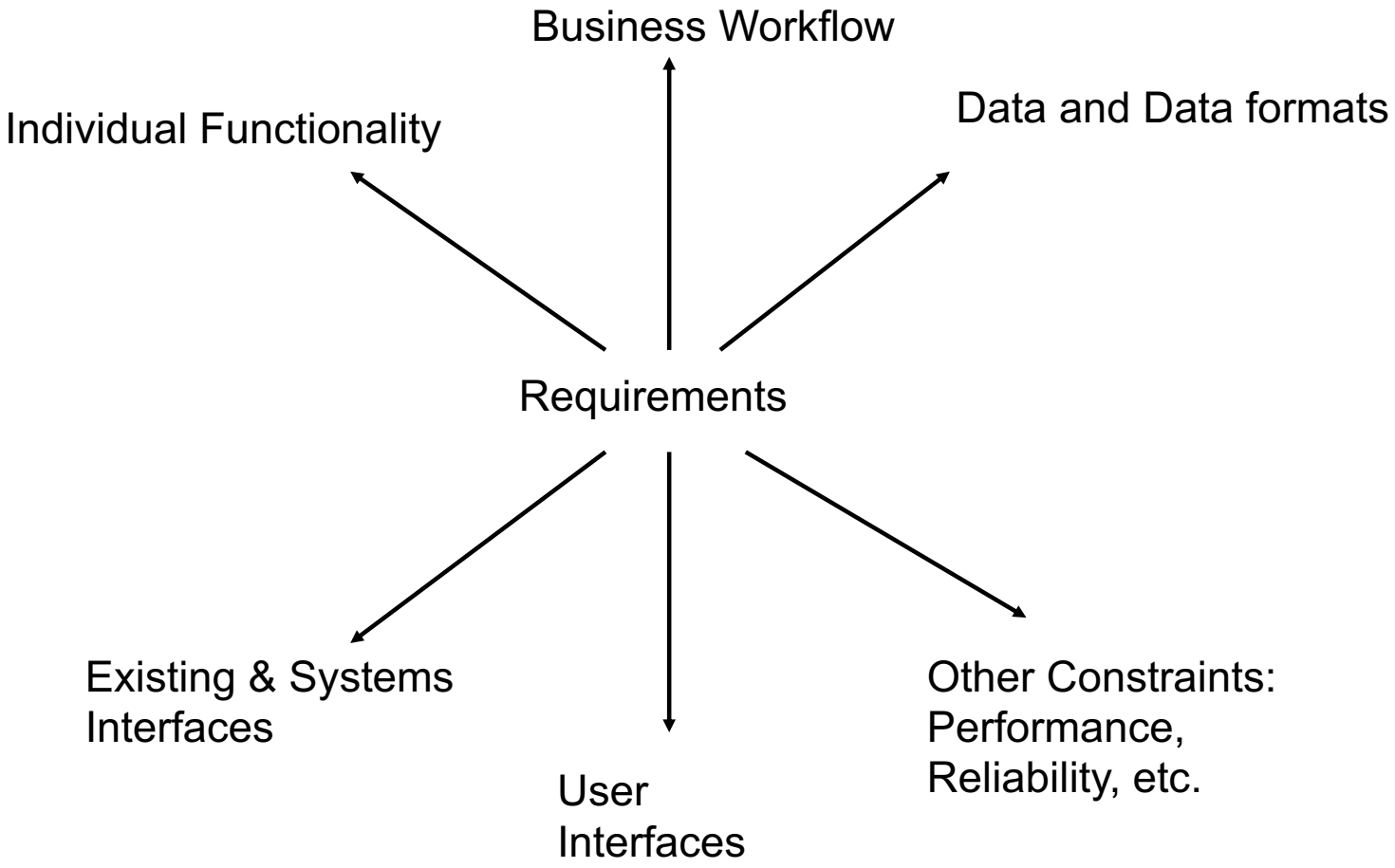


Figure 4.6 A spiral view of the requirements engineering process

6 – Dimensions of Detailed Requirements Elicitation



Requirements elicitation techniques

- It is a process of **meeting with stakeholders** of different kinds to discover information about the proposed system.
- To understand how people work, what they produce, how they use other systems, and how they may need to change to accommodate a new system.
- There are **two fundamental approaches** to requirements elicitation:
  1) **Interviewing**, where you talk to people about what they do.
  2) **Observation** or ethnography, where you watch people doing their job to see what artifacts they use, how they use them, and so on.

Interviewing

- Formal or informal interviews with system stakeholders are part of most requirements engineering processes.
- In these interviews, the requirements engineering team puts questions to stakeholders about the system that they currently use and the system to be developed.
- Two types of interview:
  - Closed interviews
  - Open interviews



Challenges/Limitations in interviews

- Stakeholders may use **jargon** specific to their area of work.
- Stakeholders are familiar with what they do so they either find it **difficult to explain** or they think it is **so fundamental** that it isn't worth mentioning.
- May not be an effective technique for eliciting knowledge about organisational requirements and constraints because there are subtle power relationships between the different people in the organization.

Tips for being an effective interviewer

Be open-minded

Avoid preconceived ideas

Willing to listen

Prompt the interviewee to get discussion going

Do not say **“tell me what you want”**



Ethnography

- Ethnography is an **observational technique** that can be used to understand operational processes and help derive requirements for software to support these processes.
- An analyst immerses himself or herself in the working environment where the system will be used, observe day-to-day work and make notes of the actual tasks in which participants (users) are involved.
- It helps discover implicit system requirements that reflect the actual ways that people work, rather than the formal processes defined by the organization.

Other requirements elicitation techniques

- Sampling of existing documentation, forms, and databases
- Research and site visits
- Questionnaires
- Joint requirements planning (JRP)

Each technique has different strengths and suitability for different scenarios

Requirements specification

Requirements specification

- It is a process of **writing down** the user and system requirements in a requirements document.
- The requirements should be **clear, unambiguous, easy to understand, complete, and consistent**.
- The requirements should be written using numbered sentences in natural language, possibly with simple tables and intuitive diagrams.
- Types of requirements specification:
  - Natural language specification
  - Structured specifications
  - Use case specification
  - Prototyping

Guidelines for writing natural language specification

1. Invent a standard format and ensure that all requirement definitions adhere to that format.
2. Use language consistently to distinguish between mandatory (must have) and desirable (nice to have) requirements.
   - "Shall" for mandatory requirements / "Should" for desirable requirements
3. Use text highlighting (bold, italic, or color) to pick out key parts of the requirement.
4. Avoid jargon, abbreviations and acronyms.
5. Whenever possible, try to associate a rationale with each requirement. The rationale should explain why the requirement has been included and who proposed the requirement.

Example of natural language specifications

- The application shall allow users to view their account balance.
- The application shall send a real-time notification to users via emails.
- The application shall require users to securely log in using multi-factor authentication (MFA).

Structured specifications (1)

No.	Field	Description
1	Requirement ID	A unique identifier assigned to each requirement for easy reference and tracking.
2	Requirement type	The category or type of the requirement, such as functional, non-functional.
3	Description	A detailed explanation of what the requirement is, outlining the specific need or functionality.
4	Rationale	The reasoning behind the requirement, explaining why it is necessary or how it supports the overall objectives.
5	Source	The origin of the requirement (who gives this requirement).

Structured specifications (2)

No.	Field	Description
6	Input(s)	Any data, information, or conditions needed for the requirement to be executed or fulfilled.
7	Output(s)	The expected result, data, or outcome produced once the requirement is implemented.
8	Action	The steps for executing this requirement.
9	Precondition	The conditions or state of the system that must be true or fulfilled before the requirement can be executed.
10	Postcondition	The conditions or state of the system after the requirement has been executed, reflecting the expected change or outcome.

Sample of a structured requirement

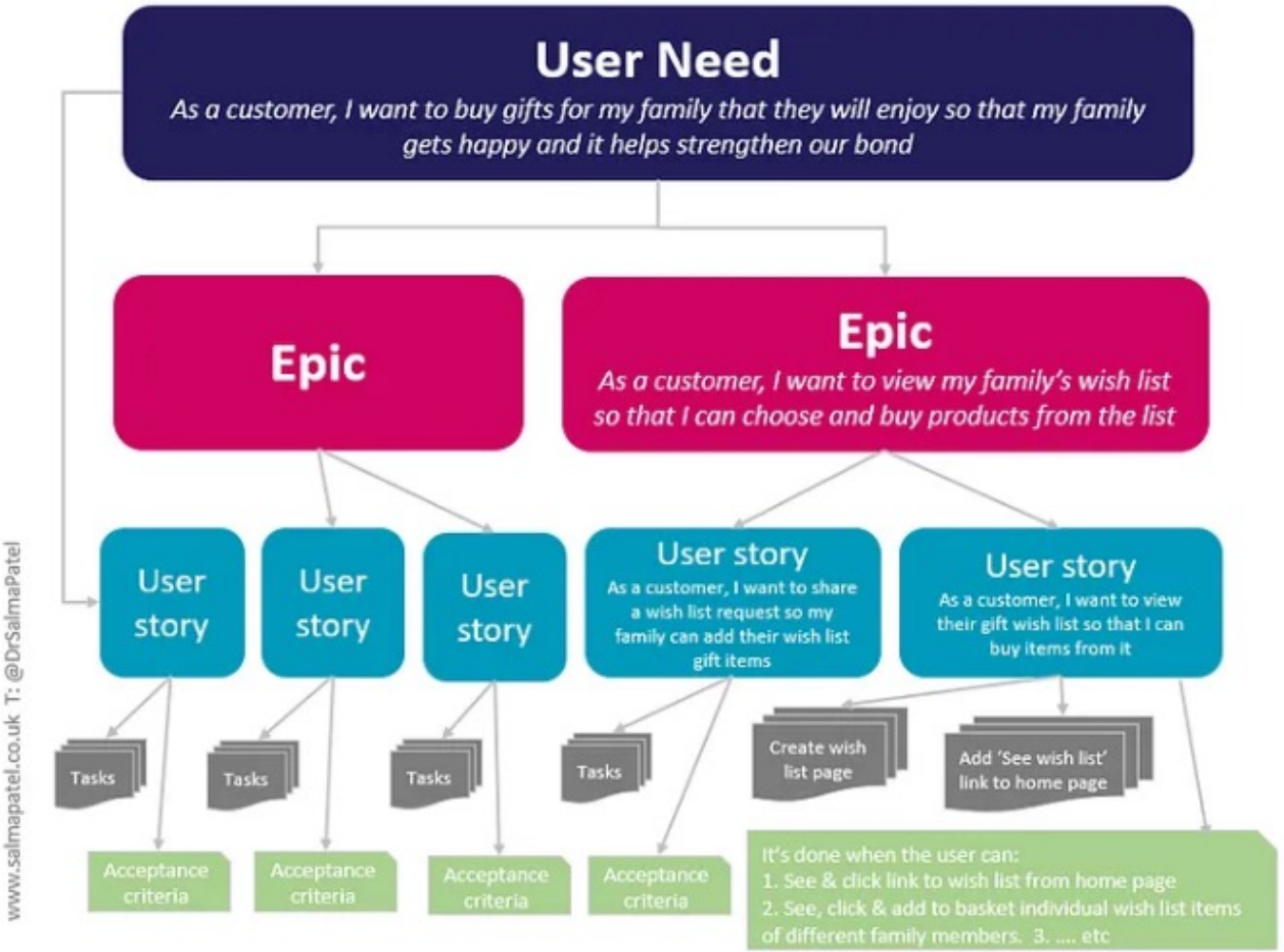
Requirement ID:	**Requirement Type:**
Description:	
Rationale:	
Source:	
Input(s):	
Output(s):	
Action:	
Precondition:	
Postcondition:	

User story

- User story is used to record requirements and define the scope of functionalities
- User story template:

ในฐานะ (As a) ...	**(บุคคล หรือ ตำแหน่ง บทบาทในระบบ)**
ฉันต้องการ (I want) ...	**(พฤติกรรม หรือความสามารถในระบบ)**
เพื่อที่ (So that) ...	**(เมื่อสร้างขึ้นแล้ว ได้ประโยชน์อย่างไร มีคุณค่าอย่างไรกับธุรกิจ)**

- User story จะถูกวัดผลด้วย “เงื่อนไขการยอมรับ (Acceptance Criteria)” หรือ Definition of Done (DoD)


User Need
As a customer, I want to buy gifts for my family that they will enjoy so that my family gets happy and it helps strengthen our bond
Epic
Epic
As a customer, I want to view my family's wish list so that I can choose and buy products from the list
User story
User story
User story
User story
As a customer, I want to share a wish list request so my family can add their wish list gift items
User story
As a customer, I want to view their gift wish list so that I can buy items from it
Tasks
Tasks
Tasks
Tasks
Create wish list page
Add 'See wish list' link to home page
Acceptance criteria
Acceptance criteria
Acceptance criteria
Acceptance criteria
It's done when the user can:
1. See & click link to wish list from home page
2. See, click & add to basket individual wish list items of different family members. 3. .... etc
www.salmapatel.co.uk T: @DrSalmaPatel

Acceptance criteria

- เงื่อนไขหรือเกณฑ์ที่ต้องทำให้สำเร็จ เพื่อให้ user story นั้นเสร็จสมบูรณ์และได้รับการยอมรับจาก Stakeholders
- คุณลักษณะของ Acceptance Criteria
  - เขียนเป็นข้อ ๆ เนื้อหาชัดเจน ไม่กำกวม
  - ต้องวัดผลได้ (measurable)
  - มักเขียนอยู่ในรูป กำหนดให้ (Given) – เมื่อ (When) – จึง (Then) (Gherkin syntax) หรือแบบ bullet point

กำหนดให้ ...	**(ผู้ใช้อยู่ในสถานการณ์อะไร)**
เมื่อ...	**(ผู้ใช้ทำอะไร)**
จึง ...	**(ผลลัพธ์ของระบบ)**

Example of user story

- **ตัวอย่างที่ 1: ระบบขายตั๋วสวนสัตว์**

ในฐานะ ผู้เข้าชมสวนสัตว์

ฉันต้องการ ซื้อตั๋วออนไลน์ผ่านเว็บไซต์

เพื่อที่ ฉันจะได้ไม่ต้องต่อคิวยาวที่หน้าทางเข้า

Example of acceptance criteria

Acceptance criterion #1

Given ผู้เข้าชมเข้าสู่หน้าเว็บขายตั๋ว

When ผู้เข้าชมเลือกจำนวนตั๋ว (ผู้ใหญ่/เด็ก)

Then ระบบแสดงราคารวมทั้งหมดให้เห็นทันที

Acceptance criterion #2

Given ผู้เข้าชมเลือกจำนวนตั๋วเรียบร้อยแล้ว

When ผู้เข้าชมทำการชำระเงินผ่านช่องทางที่รองรับ (บัตรเครดิต หรือ QR Payment)

Then ระบบยืนยันการชำระเงินสำเร็จ

Acceptance criterion #3

Given การชำระเงินสำเร็จ

When ระบบทำการออกตั๋ว

Then ผู้เข้าชมได้รับอีเมลยืนยันการซื้อตั๋ว พร้อม QR Code

Acceptance criterion #4

Given ผู้เข้าชมมี QR Code ตั๋วจากอีเมล

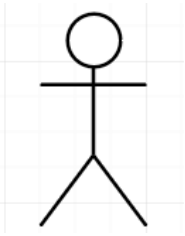
When ผู้เข้าชมไปที่ทางเข้าสวนสัตว์และสแกน QR Code

Then ระบบตรวจสอบและอนุญาตให้เข้าสวนสัตว์ได้

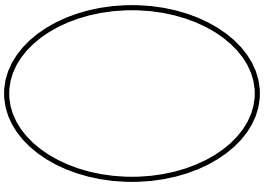
Use cases

- Use cases are a way of describing **interactions between users and a system** using a graphical model and structured text.
- The set of use cases represents all of the possible interactions that will be described in the system requirements.
- They identify the individual interactions between the system and its users or other systems.
- They are documented using a high-level use case diagram.
- Each use case should be documented with a textual description.

Use case diagram components



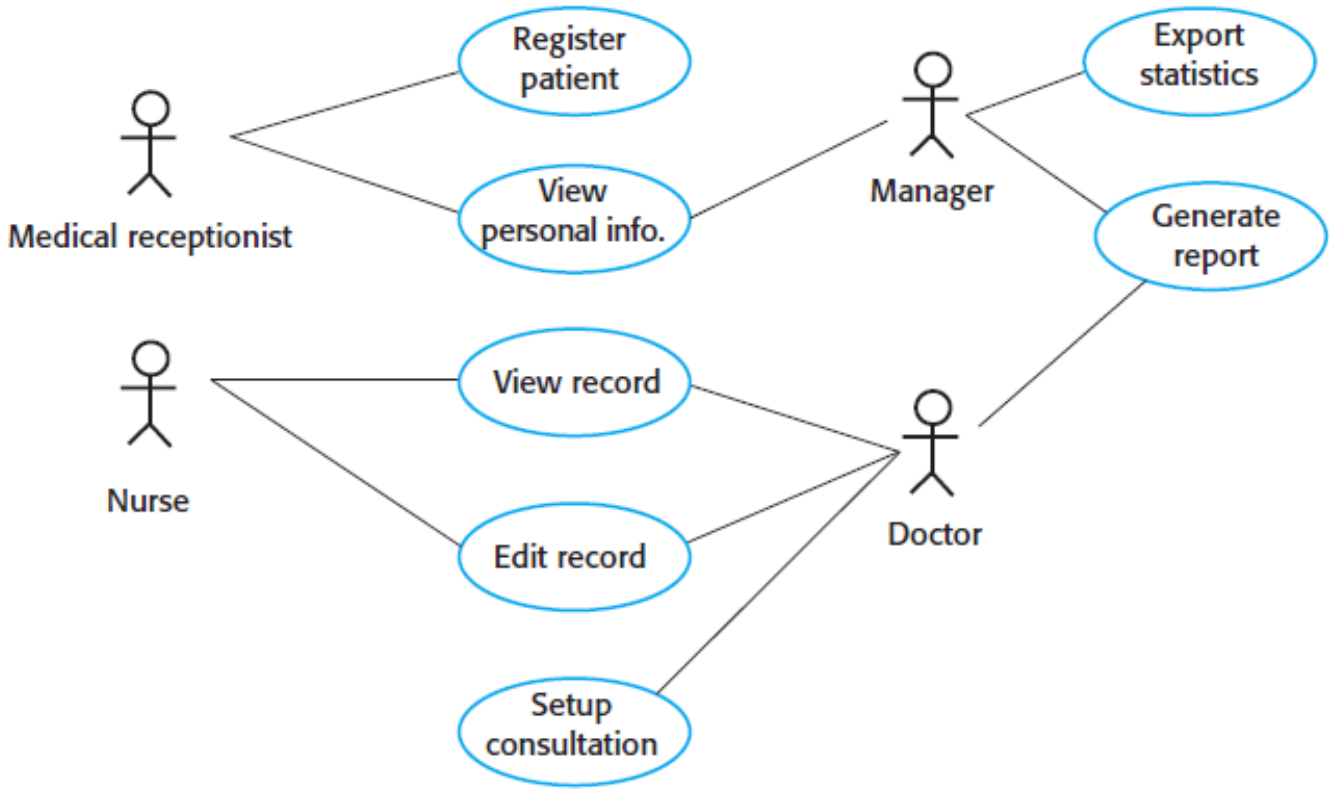
- Actors in the process, who may be human or other systems, are represented as stick figures.



- Each class of interaction is represented as a named ellipse.

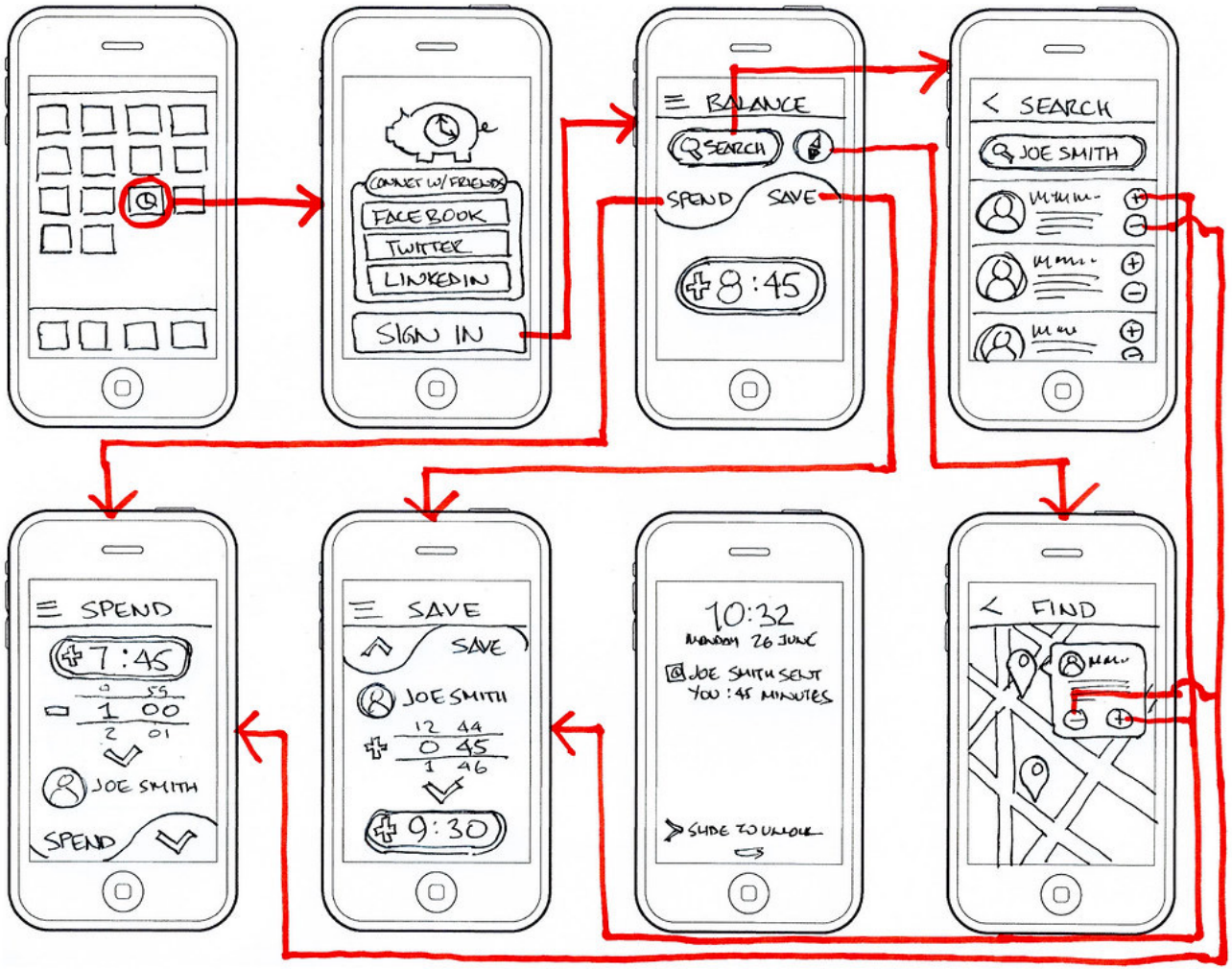
- Lines link the actors with the interaction.

Mentcare use case diagram



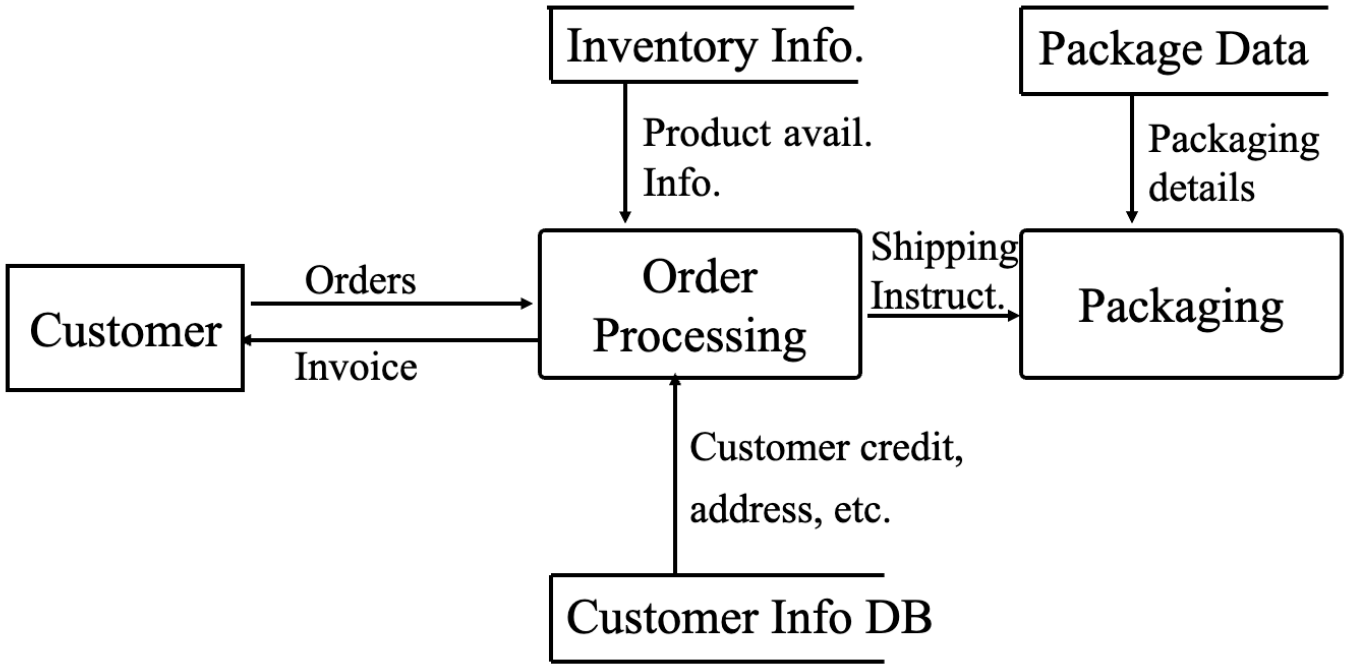
Prototyping

- Requirements prototyping mostly address the **User Interface (UI)** part of the requirement in terms of:
  - Visual looks (size, shape, position, color)
  - Flow (control and logical flow)
- The prototyping may be performed in one of the two modes:
  - **Low fidelity**: using paper/cardboard to represent screens and human to move the boards
  - **High fidelity**: using automated tools such as Visual Basic to code the screens and direct the logical flow of these screens


CONNECT W/ FRIENDS
FACEBOOK
TWITTER
LINKEDIN
SIGN IN
BALANCE
SEARCH
SPEND
SAVE
+8:45
SEARCH
JOE SMITH
SPEND
+7:45
JOE SMITH
SAVE
SAVE
JOE SMITH
+9:30
10:32
MONDAY 26 JUNE
JOE SMITH SENT YOU : 45 MINUTES
SLIDE TO UNLOCK
FIND

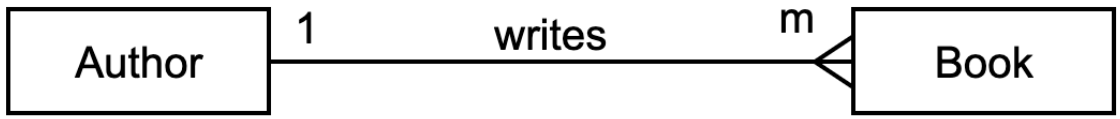
Other types of requirements specification (1)

- Data Flow Diagrams (DFD)

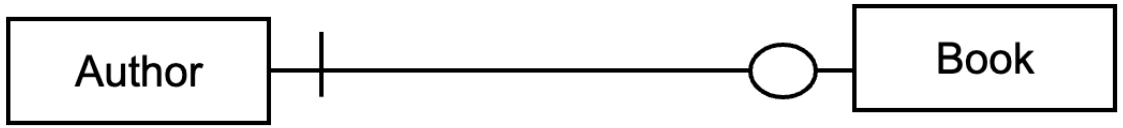


Other types of requirements specification (2)

- Entity Relation Diagram (ERD)



Cardinality: specifies the number of occurrences of entities



Modality: specifies the necessities of relationship to exist

Software requirements document

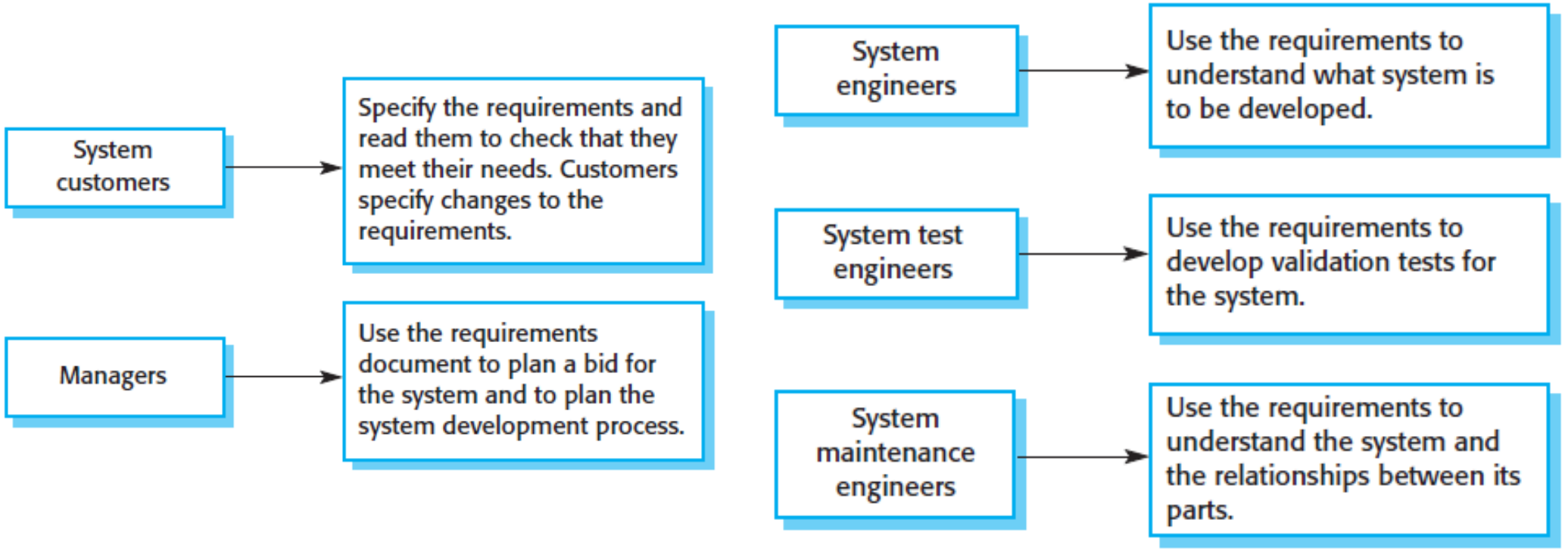
- Sometimes called the software requirements specification or **SRS**
- It is an official statement of what the system developers should implement.
- It includes both user and system requirements. However, the user requirements are described in an introductory chapter in the SRS.
- The requirements document has a diverse set of users, ranging from the senior management to the engineers responsible for developing the software.



Why requirements should be documented?

- Clear requirements are needed for **design and implementation** activities.
- Requirements documentation is needed to create **test cases and test scenarios** -- especially for large systems where the test team is a separate group of people from the developers.
- Requirements document is needed to control potential **scope creep**.
- Requirements document is needed to create **user training material, marketing material, and documents for support and maintenance**
- Requirements document provides a way to segment a large project, prioritise releases, and easier project management.

Users of the SRS



Structure of the SRS (1)

Chapter	Description
Preface	This defines the expected readership of the document and describe its version history, including a rationale for the creation of a new version and a summary of the changes made in each version.
Introduction	This describes the need for the system. It should briefly describe the system's functions and explain how it will work with other systems. It should also describe how the system fits into the overall business or strategic objectives of the organization commissioning the software.
Glossary	This defines the technical terms used in the document. You should not make assumptions about the experience or expertise of the reader.
User requirements definition	Here, you describe the services provided for the user. The nonfunctional system requirements should also be described in this section. This description may use natural language, diagrams, or other notations that are understandable to customers. Product and process standards that must be followed should be specified.
System architecture	This chapter presents a high-level overview of the anticipated system architecture, showing the distribution of functions across system modules. Architectural components that are reused should be highlighted.

Structure of the SRS (2)

Chapter	Description
System requirements specification	This describes the functional and nonfunctional requirements in more detail. If necessary, further detail may also be added to the nonfunctional requirements. Interfaces to other systems may be defined.
System models	This chapter includes graphical system models showing the relationships between the system components and the system and its environment. Examples of possible models are object models, data-flow models, or semantic data models.
System evolution	This describes the fundamental assumptions on which the system is based, and any anticipated changes due to hardware evolution, changing user needs, and so on. This section is useful for system designers as it may help them avoid design decisions that would constrain likely future changes to the system.
Appendices	These provide detailed, specific information that is related to the application being developed—for example, hardware and database descriptions. Hardware requirements define the minimal and optimal configurations for the system. Database requirements define the logical organization of the data used by the system and the relationships between data.
Index	Several indexes to the document may be included. As well as a normal alphabetic index, there may be an index of diagrams, an index of functions, and so on.

Requirements validation

Requirements validation

- It is a process of checking that requirements define the system that the **customer really wants**.
- This process is critically important because errors in a requirements document can lead to extensive rework costs when these problems are discovered during development or after the system is in service.
- A change to the requirements usually means that the system design and implementation must also be changed. Furthermore, the system must then be retested.

Requirements validation criteria (1)

- ***Validity checks:*** check that the requirements reflect the **real needs** of system users.
- ***Consistency checks:*** check that the requirements **do not have any conflicts** (no contradictory constraints or different descriptions of the same system function).
- ***Completeness checks:*** all **functions and the constraints** must be defined.
- ***Realism checks:*** to ensure that the requirements **can be implemented and delivered**.

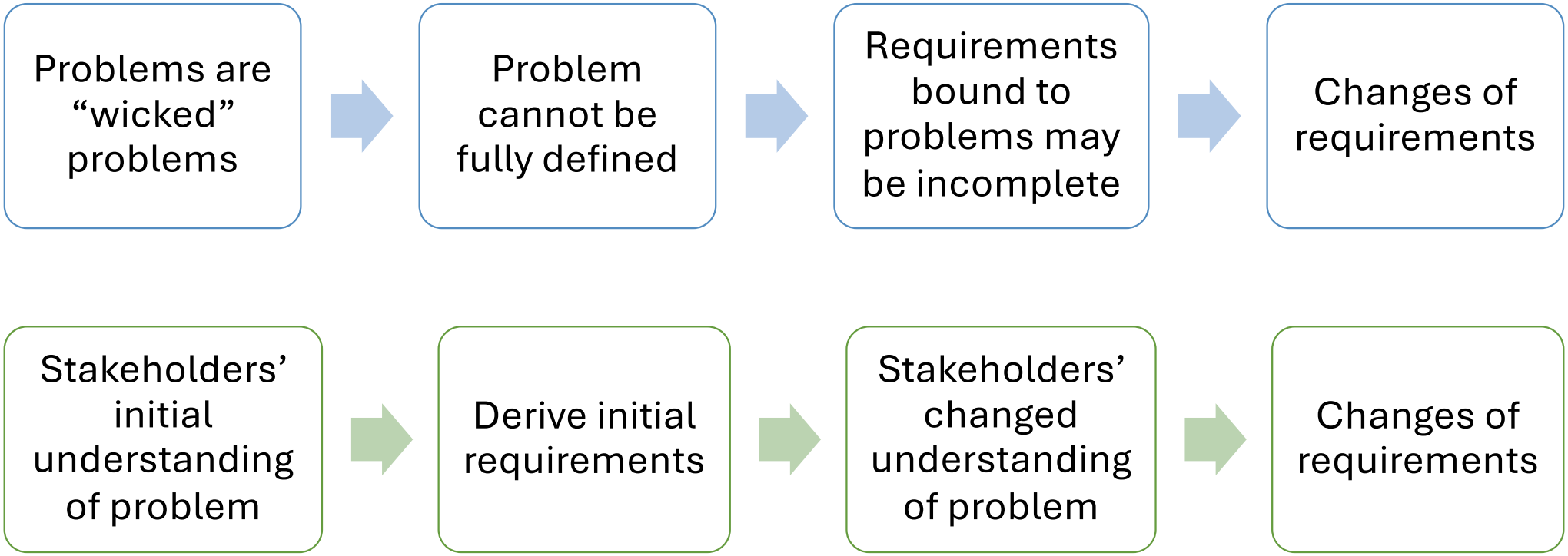
Requirements validation criteria (2)

- ***Clarity checks:*** ensure **no ambiguities** in the requirements.
- ***Verifiability:*** once the system is built, a repeatable test can be designed to demonstrate that the system fulfills the requirements
- ***Traceability:*** functional requirements correspond and can be traced back to defined system functions.



Requirements change

Requirements can change...



As requirements are evolving, you need to keep track of individual requirements and maintain links between dependent requirements so that you can assess the impact of requirements changes.
The process is called …



Requirements Management

Requirements management

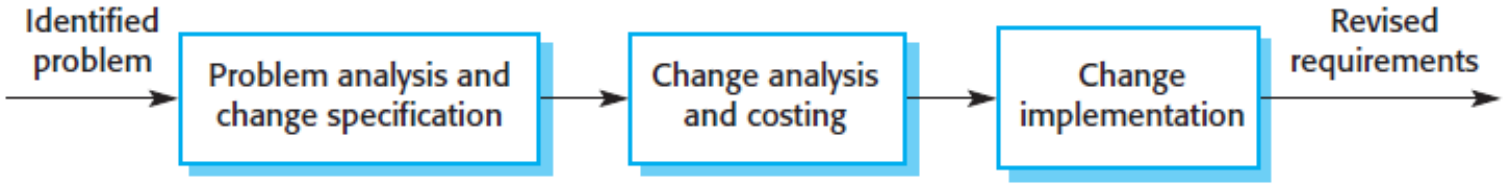
- A formal process for making change proposals and linking those changes to system requirements.
- It consists of two processes:
  - **Requirements management planning:** a process that concerns with establishing how a set of evolving requirements will be managed.
  - **Requirements change management:** a process that handles all proposed changes to a system's requirements after the requirements document has been approved.

Requirements management planning

During the planning stage, you have to consider:

- ***Requirements identification:*** each requirement must be uniquely identified so that it can be cross-referenced with other requirements and used in traceability assessments.
- ***Change management process:*** a set of activities that assess the impact of cost of changes.
- ***Traceability policies:*** define the relationships between each requirement and between the requirements and the system design.
- ***Tool support:*** an automated support for maintaining requirements.

Requirements change management



- ***Problem analysis and change specification:*** identifies a problem or change proposal, checks if it is valid, and determines acceptance or withdrawal.
- ***Change analysis and costing:*** assesses a proposed change based on traceability information and general knowledge of system requirements and estimates the cost of modifications.
- ***Change implementation:*** modifies system design and implementation.

Exercise

Given an online shopping website,

- List 5 functional requirements
- List 5 non-functional requirements